GPUscale.net v4.9.1

LLM capacity & dimensioning studio: model, precision, workload and hardware in; memory fit, latency, throughput and resilient topology out.

GPUscale.net · Untitled scenario · Kimi K3 2.8T-A50B (MoE) · weights BF16 / KV BF16 · seq 4K (+7.8K reasoning) · 250 concurrent, batch 63/replica · 16× worker (8 GPU) B300 288GB · TP32 · Active/Active · two live sites (2N) → 32 workers procured · 7/18/2026

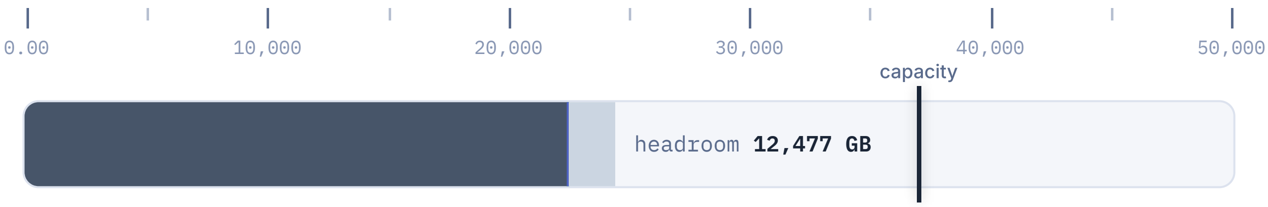
Fits, but SLO targets missed

24,387 GB required
36,864 GB across 128× GPU

Memory fits with headroom, but at least one latency/throughput target fails. See SLO compliance below.

Memory ledger

scale 0–50,000 GB · 128× B300 288GB



Weights · 4 replicas 22,400 GB 92% · KV cache · 250 seq 69.7 GB 0%

Activations 7.25 GB 0% · Overhead · fixed+multi-GPU 1,910 GB 8% · Utilization 66.2%

Time to first token

11.4 ms

prefill 4K tok on TP32 · MFU 0.5

Per-user speed

992 tok/s

ITL 1.01 ms/tok · batch/replica 62.5

Aggregate throughput

248,002 tok/s

250 admitted × per-user · 0 queued

Mean user latency

8.31 s

P95 ≈ 10.8 s · generates 8,200 tok

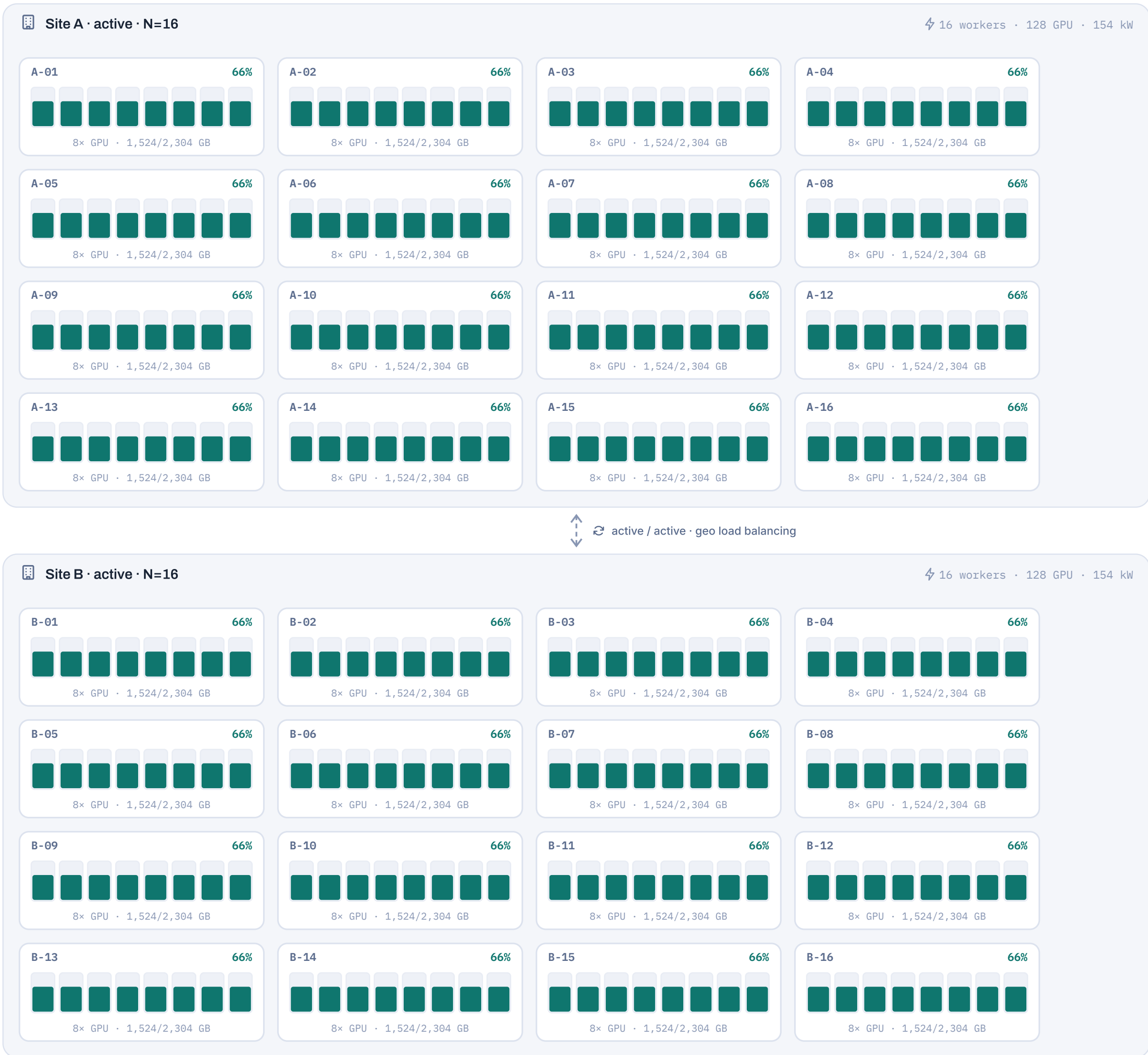
SLO compliance

targets from “Voice agent (real-time)”

TTFT	11.4 ms · ≤ 300 ms
Per-user TPS	992 tok/s · ≥ 50.0
P95 latency	10.8 s · ≤ 3.00 s

Deployment topology · workers & GPU utilization

A/A · 4 replicas × TP32 · 66%/GPU

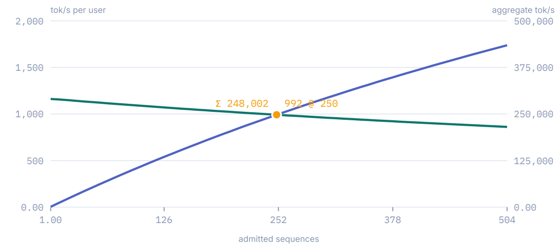


active worker · GPU bars show memory fill (66% of 288 GB each)

Load: **N = 16 workers · 128 GPUs** (8/worker): performance and fit are computed on these. Procured for Active/Active · two live sites (2N): **32 workers · 256 GPUs · ≈ 307 kW** GPU TDP. Two active sites share traffic behind global load balancing. Each site alone can carry the full load, so losing a site degrades nothing; in normal operation each runs at roughly half load.

Throughput vs admitted sequences

batch-aware



per-user tok/s aggregate tok/s

memory limit operating point

Per-user speed vs context length

at current batch

per-user tok/s current context

Request latency anatomy

mean 8.31 s · p95 10.8 s

TTFT 11.4 ms Overhead 30.0 ms Reasoning 8.06 s Visible output 202 ms

Reading of this configuration

- TP32 spans workers; the cross-node penalty is modeled via interconnect efficiency 0.7. Prefill (TTFT) is still estimated optimistically across nodes; real systems typically use TP8 inside the node plus pipeline parallelism across nodes.
- Marginal cost of one more admitted call: 0.28 GB of KV. Speculative decoding (EAGLE-3 / MTP) would add 1.5–3× on top of the decode figures.

Recommendations

no active bottleneck

TP crosses nodes (penalty modeled)

TP32 spans 4 workers and interconnect efficiency 0.7 models the cross-node cost on decode. Prefill is still estimated optimistically; production systems usually run TP8 inside the node with pipeline parallelism across nodes, or rack-scale NVLink parts.

Balanced configuration

Fits with 12,477 GB headroom (66% used), all enabled SLOs pass, and every call is admitted at peak. No action needed.

GPUscale.net

Capacity & dimensioning studio for self-hosted LLM deployments. All figures are peak estimates; production typically achieves 70–90%. Validate with vLLM bench / GenAI-Perf before commitments.

© 2026 GPUscale.net · Released under the MIT License

Version Studio 4.9.1

Engine v23 · library v24 (2026-07-18)

Scripting API at `window.GPUscale`

Private by design: everything runs in your browser. Nothing you enter is saved, uploaded or tracked. No accounts, no analytics, no cookies.